

Amiya Ranjan Kabi

Internship and Full Time Employment — 📞 07352390006 — ✉ amiyakabi28@gmail.com — Jamshedpur, India —

Profile

Innovative Electrical and Electronics Engineer with a proven track record of evaluating electrical and electronic systems and applying knowledge of electricity and technology in a beneficial way. Adept in carefully diagnosing and assessing issues, skilled in circuit designing, and testing. Committed to working as a collaborative and positive team member, striving to utilize my knowledge, previous work experience, and expertise for optimal results.

Education (Certificates)

National Institute of Technology, Jamshedpur <i>Master of Technology in Power System Engineering – 9.73 CGPA</i>	Sep 2024 – Jun 2026
Jharkhand University of Technology, Ranchi <i>Bachelor of Technology in Electrical and Electronics Engineering – 9.05 CGPA</i>	Nov 2021 – Jun 2024

Skills

Technical Skills

Electrical Systems and Maintenance

- Substation equipment and Electrical Machines, Circuit Design and Drawing Interpretation
- Power System Protection and Stability
- Power System Relaying

Process and Industrial Automation

- Plant automation using Simatic WinCC and Litmus platform using distributed control system

Digital and Embedded Systems

- IoT Integration to Electrical machines
- SAP based shift and asset management

Simulation and Design Tools

- MATLAB/Simulink
- OPAL-RT OP4510 (Real-Time Simulation and Hardware-in-the-Loop)
- Autodesk AutoCAD

Analytical and Problem-Solving Skills

- TQM and Kaizen activities in maintenance

Soft & Managerial Skills

- Team leadership and manpower management
- SOP and Energy Source Identification documentation

Projects

Tata Steel Limited, Jamshedpur

- Real-time Industrial dashboard designing using Manufacturing connect edge solution under Litmus platform and Simatic Wincc Explorer.
- Plant Automation for Sinter Furnace instruments, Pump Houses, and other areas.
- Configuration of Power Electronics Converter for Transformer rectifier panels and AC Induction motor drives.
- Commissioning of controllers for Level , moisture , temperature, pressure and other controls.
- Commissioning of distribution feeders and associated relaying controls.
- Optimised measurement of water consumption using ABB flow transmitters and controllers in entire Sinter Plant.

Project work :- Bachelor of Technology

- Weather Surveillance and Ambience monitoring Remote controlled Rover using IoT applications. (LINK)

Project work :- Diploma in Engineering

- Construction of mini hydro power plant to generate renewable electricity.

Research Publications

National Institute of Technology, Jamshedpur

Oct 2024 – Present

- **Kabi, A. R.**, et al., “Comparative Analysis between Battery Energy Storage and Hydrogen-Based Fuel Cell Energy Storage for Rooftop PV Systems,” *IEEE SEFET 2025*, MNIT Jaipur, June 2025. Published in IEEE Xplore.
- **Kabi, A. R.**, et al., “Estimation and Comparison of Monthly Average Daily Global Solar Radiation Using Empirical Models,” *IEEE EPREC 2026*, IIT Bhilai, January 2026. Published in IEEE Xplore.
- **Kabi, A. R.**, et al., “Energy Management and Electrical Stress Analysis of Grid-Tied Rooftop PV with Hybrid Lithium-Ion Battery and Reversible PEM Storage Systems,” Extended version of SEFET 2025 paper. Submitted to *IEEE Transactions on Sustainable Energy*, 2026. Under review.
- **Kabi, A. R.**, et al., “Development of a Bidirectional Fuel Cell–Electrolyzer Integration Block in MATLAB/Simulink for Distributed Energy Systems,” *International Journal of Modelling and Simulation* (Taylor & Francis). Under peer review.
- **Kabi, A. R.**, et al., “SoH-Aware Adaptive Fast Charging Strategy for Electric Vehicle Batteries,” International Conference, NIT Meghalaya, June 2026. Submitted.

Courses and Certifications

Tata Steel Limited

- School of Excellence - Electrical Motor and Protection
- School of Excellence - Industrial Internet of Things
- School of Excellence - Low Voltage AC Drives
- School of Excellence - PLC and Automation
- School of Excellence - Process Instrumentation and Control System

Extra Curricular Activities

Volunteering in Jamshedpur Blood Bank

Joint Development Committee Blood donation Camps

Member of NGO, Green Nation

Feb 2022 – Jun 2024

Green nation does field work to support the nearby villagers with education, healthcare and livelihood assistance.

Hobbies

- **Guitar Learner**
- **Free Style Swimming**
- **Outdoor Activities**